Interspeech 2026
27 September - 1 October 2026, Sydney, Australia

Rethinking Acoustic Variability Of ADReSS and ADReSSo Datasets For Dementia Detection

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Abstract

The ADReSS and ADReSSo challenge datasets have become the de-facto standards for research on dementia detection through speech, with over half of recent ICASSP and Interspeech studies on the topic relying on them. Despite their widespread adoption, both datasets exhibit properties that may undermine the validity of reported results. In this paper, the acoustic variability in the test sets of ADReSS and ADReSSo is systematically examined. Results show that (i) near state-of-the-art classification performance on the test sets can be achieved using only two low-level acoustic features, (ii) classifiers trained on randomly permuted dementia labels can nonetheless reach competitive test performance, and (iii) feature discriminability does not generalize across Monte Carlo resampling, with no stable features emerging from low-level acoustic feature sets. These findings indicate that strong results on these datasets can stem from spurious correlations rather than pathology-relevant cues. The analysis calls for more cautious interpretation of benchmark performance, as well as renewed attention to small dataset evaluation strategies in dementia detection.

Index Terms: Alzheimer, dementia, cognitive impairment, ADReSS, ADReSSo, Pitt corpus, acoustic LLDs, openSMILE

1. Introduction

Dementia is a progressive neurodegenerative condition affecting millions worldwide, with incidence projected to triple by 2050 [1]. Early diagnosis is critical, as both pharmacological and lifestyle interventions have shown promise in slowing disease progression [2]. Cognitive decline associated with dementia disrupts multiple processes, including memory, language, and executive function, many of which manifest in spontaneous speech [3]. This has motivated growing interest in speech-based approaches to dementia detection, leading to the release of multiple benchmark datasets through community challenges [4, 5, 6, 7, 8]. Among these, the ADReSS [4] and ADReSSo [5] challenges stand out as the most widely adopted resources, featuring American English speech samples of dementia and healthy control participants.

Since 2020, more than 100 papers in the ICASSP and Interspeech proceedings alone have addressed dementia or cognitive decline detection using speech, with over half reporting classification results using ADReSS or ADReSSo datasets. These datasets have thus become the de facto standard for speech-based dementia research. However, despite their widespread use, both datasets present challenges that remain underexamined, notably substantial acoustic variability and optimistic test set behavior. This paper revisits these issues and highlights their implications for model evaluation and generalization.

The ADReSS and ADReSSo datasets are derived from the Pitt Cookie Theft Corpus [9], which has since been shown to permit trivially easy discrimination between dementia and healthy speech using only silent segments [10]. To mitigate such confounds, the ADReSS and ADReSSo challenge organizers applied preprocessing steps intended to normalize recording conditions across the datasets. Nevertheless, residual imbalances remain, raising concerns about whether models trained on these datasets capture pathology-relevant cues or instead exploit incidental acoustic artifacts. Subsequent studies have highlighted related issues, including the influence of the interviewer's speech in distinguishing dementia from healthy speech [11] and achieving above-chance discriminability using silent segments in the ADReSSo dataset [12].

While these studies have established that particular segments of the recordings can produce inflated performance, the question of whether the entire recording is susceptible to brittle classification on low-level acoustic features remains underexplored. To address this question, we re-examine the ADReSS and ADReSSo datasets and present three key observations:

- Near state-of-the-art classification performance can be obtained with just two low-level acoustic features on the challenge test sets. Closer inspection reveals that these features may encode little beyond channel-related variability.
- Even when dementia labels are randomly permuted (some control recordings re-labelled as dementia and vice versa), classifiers may achieve competitive test set performance with just two low-level acoustic features.
- The test set performance does not generalize across Monte Carlo resampling splits of the datasets. While subsets of acoustic features discriminate individual splits, no feature generalizes reliably in either feature sets investigated.

These findings suggest that strong classification performance on ADReSS and ADReSSo may arise from spurious correlations specific to the challenge split, calling into question the robustness and generalizability of systems evaluated on these benchmarks. The results highlight the need for more cautious interpretation of benchmark performance and for improved evaluation practices when working with small dementia speech datasets.

2. Methodology

The stability of low-level acoustic features on the ADReSS and ADReSSo dementia detection datasets is evaluated using an interpretable classification pipeline. Acoustic descriptors from the openSMILE toolkit [13] are extracted and assessed in pair-

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10.21437/Interspeech.2026-2862

wise combinations using logistic regression classifiers. Performance is examined under multiple evaluation regimes, including the original challenge test split, label permutation, and Monte Carlo resampling. In addition, speech-only and silence-only dataset variants are constructed using voice activity detection to consolidate the hypothesis that high-performing features may not be reliably linked to speech cues.

2.1. Datasets

Two benchmark datasets are studied: the ADReSS dataset released in 2020 and the ADReSSo dataset released in 2021, both of which provide balanced subsets of speech recordings from dementia patients and healthy controls [4, 5]. The speech consists of participants performing the Cookie Theft picture description task [14], in which individuals describe a complex kitchen scene illustrated in a static drawing. Both ADReSS and ADReSSo are derived from the Pitt Corpus, but with additional preprocessing steps. Crucially, both datasets are pre-balanced by age, gender, and diagnosis to control for potential confounds. The dataset composition is summarized in Table 1.

Table 1: *Dataset composition of ADReSS and ADReSSo, showing the number of healthy control (HC) and Alzheimer's dementia (AD) participants in the train and test sets.*

Dataset	Split	HC	AD
ADReSS	Train	54	54
	Test	24	24
ADReSSo	Train	87	79
	Test	36	35

The pyannote [15] Voice Activity Detection (VAD) library is used to segment each audio recording into speech and non-speech regions, from which two derived dataset variants are constructed. The first is a silence-only variant, where each recording consists of a concatenation of all silence segments from the original audio. The second is a speech-only variant, where each recording is formed by concatenating only the detected speech segments.

2.2. Feature Sets

Low-level acoustic descriptors (LLDs) have been used as baseline models in all five dementia detection challenges to date. These descriptors are well-curated in the openSMILE toolkit [13], with two of the most prominent sets being the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [16] and the Computational Paralinguistics Challenge feature set (ComParE) [17]. The eGeMAPS feature set (`eGeMAPSv02`) comprises 88 features and provides a compact, theory-driven summary of prosodic, spectral, and voice-quality characteristics. By contrast, the ComParE feature set (`ComParE_2016`) is far more exhaustive, comprising 6,373 features that capture a wide spectrum of prosodic, spectral, energy-based, and cepstral descriptors, along with extensive statistical functionals.

2.3. Feature Selection

Given the large number of descriptors in both eGeMAPS (88 features) and ComParE (6,373 features), a stepwise feature selection strategy is adopted. First, classifiers to discriminate control from dementia recordings are trained on each individual feature in isolation, allowing us to quantify its discriminative power on both the fixed test set and on resampled splits generated during the Monte Carlo validation (see Section Validation). From this analysis, the top-performing individual features are identified and in a second stage, pairwise combinations of these top features are used to train classifiers. This allows for the investigation of whether complementary information improves classifier performance. The minimal feature pairing is a deliberate attempt to examine how far classifier performance can be pushed by exploiting a small subset of features.

2.4. Validation

Logistic regression classifiers are used in all experiments, as they provide an interpretable linear decision boundary in the 2D case. By design, if logistic regression achieves meaningful separation between dementia and healthy speech, then more complex models are unnecessary.

The implementation follows a standard `scikit-learn` pipeline. Missing values are imputed with the mean, and all features are z-score normalized. The `liblinear` solver is used due to its stability in high-dimensional, small-sample settings.

All classifier performance is reported using the macro-F1 score, which provides an equal-weighted balance of precision and recall across both classes. This ensures that classification quality is not biased toward the majority or easier-to-classify class.

Label Permutation: To evaluate the robustness of the classification, label permutation experiments are conducted as a negative control. For each of 100 iterations, the class labels in the training split - *dementia* and *control* - are randomly permuted and the classifier is retrained and evaluated. The resulting performance represents the chance-level baseline of the pipeline under label-shuffling. This procedure ensures that any observed discriminative power in the real experiments is not an artefact of noise or spurious correlation.

Monte Carlo Resampling: To evaluate the generalizability of the features, Monte Carlo resampling validation is conducted. Specifically, we repeatedly sample random train/test splits in a 70/30 ratio from the full dataset while ensuring that pathology labels remain balanced across splits. This ratio mirrors the composition of original datasets. Each experiment is repeated for 100 iterations, allowing us to average over different distributions of age, gender, and MMSE.

3. Results

The results are organized around three key observations as follows:

3.1. Optimistic Test Sets

The challenge test sets of both ADReSS and ADReSSo are considered optimistic in the sense that good classifier performance is reached with only two low-level acoustic features. In particular, classifier performances close to competition state of the art are obtained, as given in Table 2.

Table 2: *Comparison of two-feature classification results against competition state-of-the-art (SOTA) benchmarks on ADReSS and ADReSSo. All numbers are macro F1.*

Dataset	2 Features	SOTA
ADReSS	0.875	0.895 [18]
ADReSSo	0.831	0.857 [11]

Table 3: *Two-feature pairings used for classifying ADReSS and ADReSSo test sets. The names are from openSMILE.*

Dataset	Two features per classifier
ADReSS	1. `audSpec_Rfilt_sma[22]_quartile1`, 2. `pcm_fftMag_spectralCentroid_sma_de_quartile2`
ADReSSo	1. `mfcc_sma[6]_centroid`, 2. `mfcc_sma[11]_flatness`

The features responsible for these results are given in Table 3.

Figure 1 illustrates the decision boundaries obtained with these features on train and test splits. Despite the clean separation implied by the decision boundaries in the test split, the features themselves are not strongly separable in the train split, as shown in Figure 1.

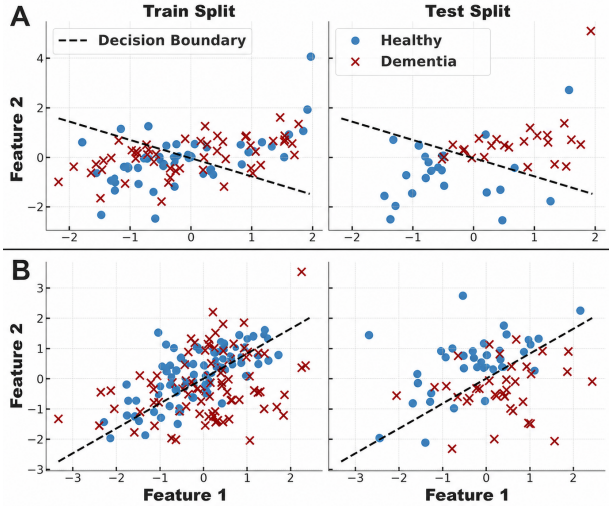


Figure 1: *Logistic regression decision boundaries for A) ADReSS and B) ADReSSo datasets using two acoustic features listed in Table 3. Left: train split. Right: test split.*

From an acoustic perspective, the selected features merit closer examination. For ADReSS, `audSpec_Rfilt_sma[22]_quartile1` represents the first quartile of a band-limited auditory spectrum filter, effectively a summary statistic of energy distribution in one subband. This measure is susceptible to recording channel characteristics and background noise. By contrast, `pcm_fftMag_spectralCentroid_sma_de_quartile2` captures the median value of the derivative of the spectral centroid, i.e., the distribution of energy across frequencies over time. While spectral centroid dynamics may plausibly relate to articulatory or phonatory changes, the feature remains sensitive to artefacts such as microphone response and silence handling.

For ADReSSo, `mfcc_sma[6]_centroid` is the centroid statistic of the sixth Mel-frequency cepstral coefficient. This coefficient primarily reflects spectral envelope shape and may react to duration and voice activity detection boundaries. As such, it is prone to channel confounds. The second feature, `mfcc_sma[11]_flatness`, measures the flatness of the eleventh MFCC trajectory, which can correspond to reduced articulatory variability. Nevertheless, it too remains vulnerable to stationary noise and recording-chain effects.

Empirical evidence that the selected features are encoding channel-specific and not speech-specific cues is shown in Table 4. Classifiers trained on VAD-extracted silence-only and speech-only variants of the datasets show that the silence-only datasets outperform the speech-only datasets on the same two-feature combinations. This indicates that these features are more discriminative within the silent segments for both ADReSS and ADReSSo, and can not be reliably linked to speech cues.

Table 4: *Comparison of silence-only and speech-only recordings using features from Table 3. All numbers are macro F1.*

Dataset	Silence-Only	Speech-Only
ADReSS	0.702	0.643
ADReSSo	0.631	0.576

3.2. Convincing Random Labels

Even when dementia labels are randomly reassigned within the training set, classifiers trained on just two low-level acoustic features achieve test set scores that overlap with the best test set performance.

Specifically, across 100 repeat label permutations of the train set labels, the two-feature models produced a wide spread of test performance. For ADReSS, macro F1 scores ranged from 0.123 to 0.875, and for ADReSSo from 0.153 to 0.831. These distributions are shown in Figure 2, where each point represents the outcome of a single permutation run.

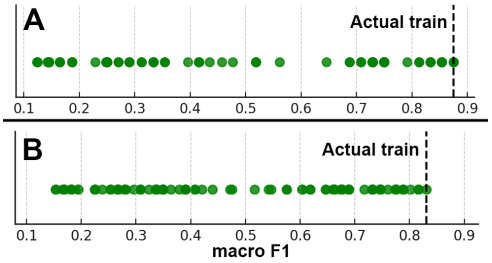


Figure 2: *Scatter plots of performance under label permutation for A) ADReSS and B) ADReSSo. Each dot corresponds to a random permutation of the training labels. Dashed vertical lines indicate the benchmark test scores reported in Table 2.*

As Figure 2 illustrates, there exist non-negligible cases of random labeling where the classifier attains scores comparable to those reported with actual labels. That is, chance configurations of spurious correlations between features and labels can occasionally produce results that are deceptively strong.

3.3. Unstable High Performing Features

Although individual test splits suggest that certain combinations of features yield high scores, their performance does not generalize across resampling.

First, the same two-feature pairs that achieved near state-of-the-art performance on the challenge test sets are evaluated by examining their performance across 100 Monte Carlo resampling splits. Figure 3 shows the resulting distribution of

macro F1 scores for ADReSS and ADReSSo. Whereas the test set scores approached 0.875 and 0.831 respectively (see Table 2), the Monte Carlo mean values drop substantially: 0.622 for ADReSS and 0.667 for ADReSSo. This gap illustrates that the apparent strength of the features on a single benchmark split does not translate into stable discriminative power across different data partitions.

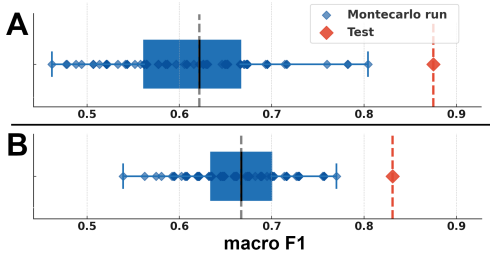


Figure 3: *Monte Carlo distributions of macro F1 scores for the two-feature models on A) ADReSS and B) ADReSSo. Each box summarizes 100 resampling splits. Dashed lines indicate the respective resampling means and benchmark test set scores.*

Furthermore, the emergence of consistent feature pairs across Monte Carlo iterations is investigated. For each of the 100 iterations, the 25 highest-performing two-feature combinations are recorded. The resulting sets are then compared across iterations to determine how frequently a given feature pair recurs. Figure 4 plots a histogram of these counts, showing the number of feature pairs that occur in at least x iterations. For feature pairs from eGeMAPS, only 3 and 1 feature pairs recur in at least 50% of the iterations in ADReSS and ADReSSo respectively, while no feature pair recurs 50% of the time from the ComParE set.

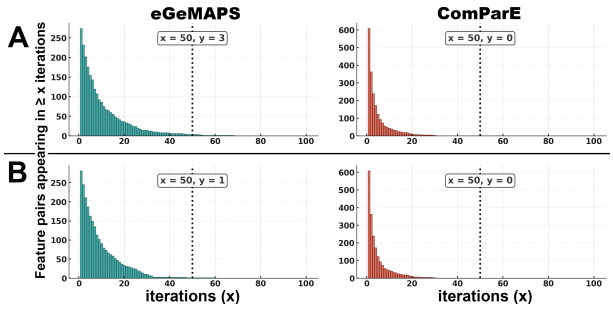


Figure 4: *Histogram of feature recurrence across 100 Monte Carlo runs on the eGeMAPS and ComParE features across A) ADReSS and B) ADReSSo datasets. Each bar indicates the number of feature pairs appearing in at least x iterations.*

The feature pairs that recur in more than 50% of runs for the eGeMAPS feature set are listed in Table 5. These pairs achieve macro-F1 scores of 0.729 (1A,1B), 0.646 (2A,2B), and 0.624 (3A,3B) on the ADReSS dataset, and 0.690 (1A,1B) on ADReSSo, following the ordering in Table 5. Notably, even the most frequently occurring high-performing pairs do not exceed 0.73 on ADReSS or 0.69 on ADReSSo. Thus, despite thousands of feature combinations evaluated across runs, none demonstrate the level of stability expected of robust, generalizable discriminative features.

Table 5: *Two-feature pairs recurring in $\geq$ 50% of Monte Carlo splits from eGeMAPS. The names are from openSMILE.*

Dataset	**Feature pairs recurring in $\geq$ 50% of MC splits (eGeMAPS)**

ADReSS	`1A. F0semitoneFrom27.5Hz_sma3nz_percentile80.0`
	`1B. loudness_sma3_percentile50.0`
	
	`2A. F0semitoneFrom27.5Hz_sma3nz_amean`
	`2B. loudness_sma3_percentile50.0`
	
	`3A. F0semitoneFrom27.5Hz_sma3nz_percentile80.0`
	`3B. loudness_sma3_percentile20.0`

ADReSSo	`1A. F0semitoneFrom27.5Hz_sma3nz_percentile80.0` `1B. VoicedSegmentsPerSec`

4. Discussion

Results suggest that strong scores on the fixed ADReSS ans ADReSSo test splits can arise without stable, pathology-linked signal. Two-feature models reached high macro F1 on the benchmark tests (0.875 and 0.831; Table 2), yet their mean performance under Monte Carlo resampling dropped markedly (0.622 for ADReSS; 0.667 for ADReSSo; Fig. 3). Moreover, label-permutation controls occasionally overlapped the true-label test scores (Fig. 2), and no feature combination generalizes well across Monte Carlo splits. These observations indicate that optimistic test results are possible when small splits align with incidental structure in low-level descriptors, rather than reflecting robust dementia-related cues. The drop in performance when looking at the silence-only and speech-only metrics suggests that these features are quite brittle.

These findings are read as a caution about evaluation, not as a blanket indictment of prior work. First, the acoustic feature families we explored exhibit high within-set variability: many descriptors are sensitive to channel and background conditions, and pairwise analyses showed little recurrence of "winning" features across resamples (Fig. 4). Second, the dataset sizes are modest, so a few influential points can shift linear boundaries on a single split. This might explain why test-set gains did not translate into generalization.

Practically, this argues for more robust validation protocols. Beyond reporting a single test score, studies should (i) run repeated, stratified resampling or nested cross-validation and report dispersion (e.g., Monte Carlo mean and variance); (ii) include negative controls (e.g., label permutation) to calibrate chance alignment; and (iii) treat results on a prescribed split as one estimate among many.

Methodologically, the findings underscore the value of models whose behaviour is explainable and anchored in domain knowledge. Low-level acoustic descriptors are useful probes, but robust systems should connect decisions to clinical markers in dementia and make those links inspectable. In short, careful validation and explainable modeling are both needed to turn promising test-set performance into trustworthy evidence for dementia detection.

5. Acknowledgement

This work was supported by the National Institutes of Health (NIH) under Grant No. 5R01AG057531.

6. Generative AI Use Disclosure

A Generative AI tool was used to proofread this manuscript and to enhance the visual presentation of figures. After using the tool, the authors reviewed and edited the content as needed and take complete responsibility for it.

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